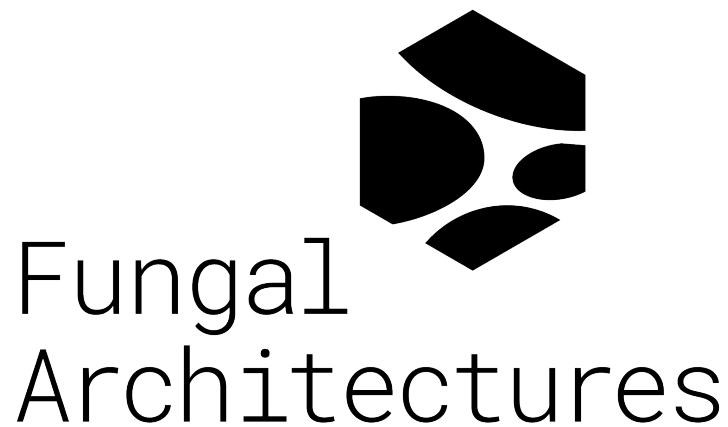


Horizon 2020



Deliverable D5.3

Report on design rules for fungal architecture

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Project coordinator: UWE	Classification: public
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DELIVERABLE SUMMARY SHEET

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Name: **Report on design rules for fungal architecture**

Description: In this deliverable, we report on activities and results related to T5.3 *Design rules for fungal architecture*. This includes investigating and defining novel architectural design rules that integrate spatial organisation, structural requirements and computational functions at architectural scale. The topology and geometry of designs will be investigated as the principle control parameters for achieving integration. This will be investigated in simulation (using established generative and analytic parametric modelling methods) and physical prototypes.

Partners owning: **CITA**

Partners contributed: **CITA**

Made available to: public

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1 INTRODUCTION

In this deliverable we report upon the definition of design rules for Fungal Architectures. These rules target two key objectives that seek to offer a solution to the problem statement defined in the project abstract:

As one of the primary consumers of environmental resource, the building industry faces unprecedented challenges in needing to reduce the environmental impact of current consumption practices. This applies to both the construction of the built environment and resource consumption during its occupation and use. Where incremental improvements to current practices can be realised, the net benefits are often far outstripped by the burgeoning demands of rapidly increasing population growth and urbanisation. Against the backdrop of this grand societal challenge, it is necessary to explore approaches that envision a paradigm shift in how material is sourced, processed and assembled to address the magnitude of these challenges in a truly sustainable way, and which can even provide added value.

1. that the design rules embody an inclusive, equitability, affordable, sustainable construction approach with low barriers to entry and built upon principles of ecological and social justice, community practice and empowerment through accessible craft mastery.
2. that the rules lead to unique spatial, tectonic and architectural expressions - building upon characteristics established by the first construction concept prototype (shown previously at the start of section T5.1 - Growth Scaffolds).

We have approached the first objective by developing a construction concept that combines two of humanity's most enduring technologies - weaving and fermentation. The pan-cultural Kagome weave technique is over 8000 years old. Weaves can be made from an abundant palette of materials and the technique can be quickly learned. Combining this with solid-state-fermentation techniques that can exploit ligno-cellulosic waste streams to create viable and performing material composites, provides the basis for a novel and accessible approach to construction.

In approaching the second objective, particular attention has been paid to developing a digital design workflow to allow open design exploration with specific focus on weave pattern topology, morphology, structural performance and spatial organisation.

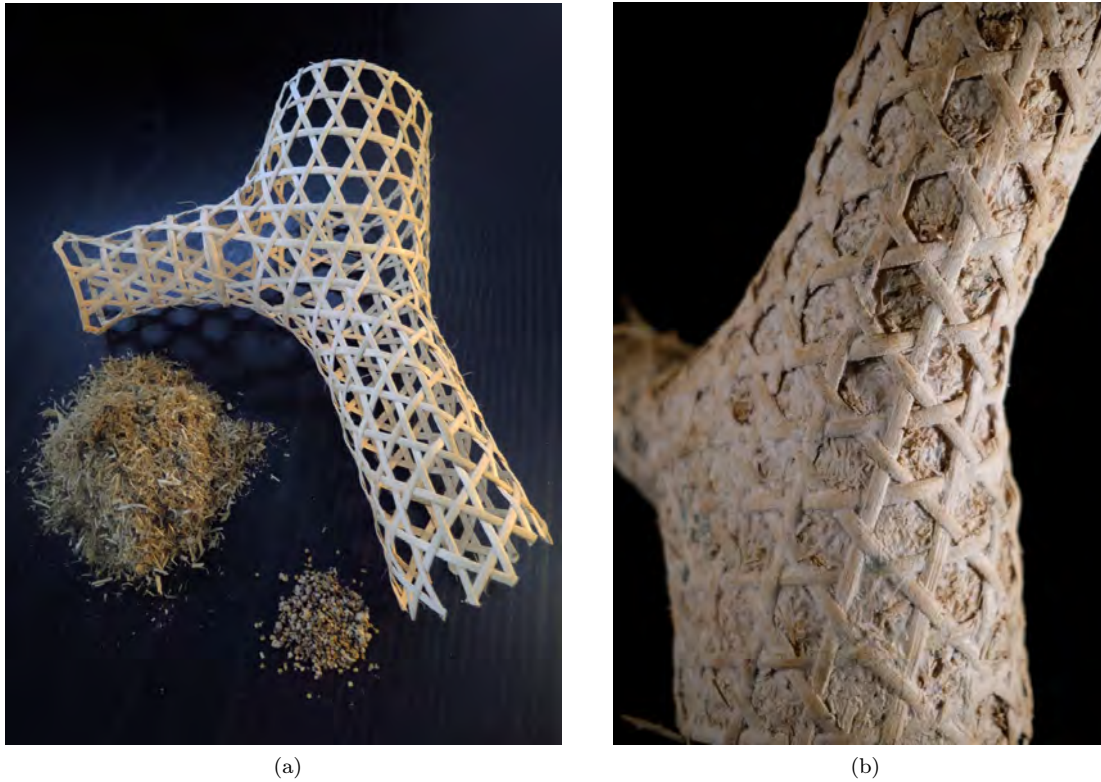


Figure 1: Initial prototype of a novel construction concept for Fungal Architectures, produced for the grant application (2018). Constituents (a) and the grown prototype (b)

2 FOCUS AREAS OF DESIGN DEVELOPMENT

The design rules and workflow presented in Section 3 - Design Rules, reflect significant revisions to the original construction concept (Fig.1). These revisions have been informed by the many experimental findings, empirical investigations of mycelium-based composite (MBC) mechanical properties and 1:1 prototyping of assemblies (reported on in Deliverable D2.3) conducted across the project period. Principal revisions to the original concept and objectives include:

1. the introduction of an independent Kagome gridshell acting as primary structural support for an MBC enclosure
2. falsifying the notion of macro-scale morphological adaptations through mycelium growth
3. falsifying the notion of occupiable spaces being composed of monolithic living mycelium
4. needing to contain living mycelium in distinct and differentiated conditions to human occupants
5. the use of a single kagome woven surface for the application of MBC as opposed to a ‘filled Kagome’ approach
6. the use of denatured MBC moulded panels, without Kagome weaves, to act as spatial separators within the domain of the gridshell enabling the consideration of component elements to spatial organisation

In this section, we briefly report upon the four core areas of investigation that have informed the architectural development of the construction concept. These are, the Kagome modelling approach, limits in mechanical performance of MBC, Kagome weaving with solid timber members and broader design speculations that guide the development of the spatial vocabulary for Fungal Architectures.

2.1 Kagome Modelling

In deliverable D5.1 - Growth Scaffolds, we reported upon the development of the Mesh Topology Adjustment Scheme (MTAS) - a method for designing principled Kagome weave patterns using regular computational tri-meshes as a basis [1]. Our subsequent investigations in Kagome modelling have developed a more generalised approach that overcomes a number of restrictions of MTAS. These included:

1. manual locating of vertices for topology adjustment
2. sequences of singularity placement could result failed topology designs despite being physically weavable
3. closed geometries failing due to circular dependencies in topology adjustment
4. a regular planar tri-mesh as a starting point required a strong design intuition to produce complex weave pattern designs that approximate desired morphologies

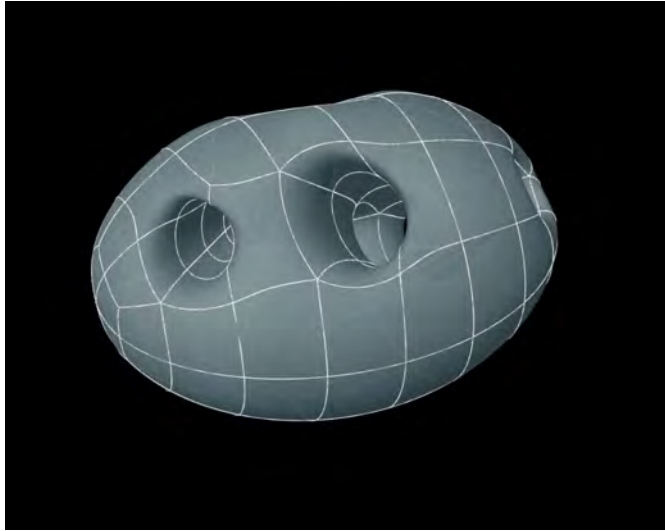


Figure 2: Kagome modelling, step 1: defining a design target using sub-d modelling

To address these constraints, we developed a new method that begins with the design of the desired morphology using the designer's preferred modelling method - in the case shown here (Fig.2), using a Sub-D modelling approach.

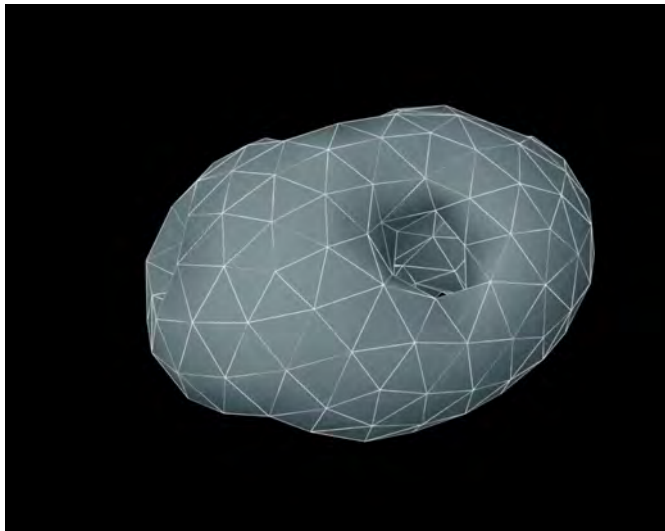


Figure 3: Kagome modelling, step 2: meshing with fixed edge length constraint

The Sub-D model is then re-meshed using an iterative constraint-based approach with a specified target edge length for the mesh-faces that corresponds to the weave cell size. This re-meshing results in network of equilateral triangles that embed a principled topology corresponding to the necessary placement of singularities in the weave (Fig.3).

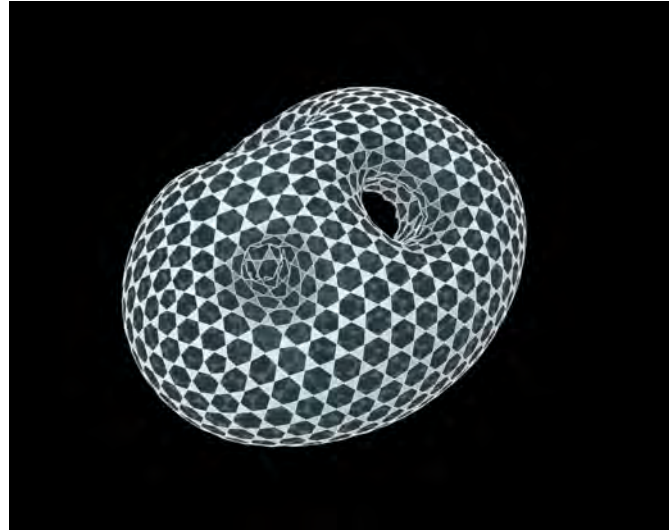


Figure 4: Kagome modelling, step 3: medial construction of kagome pattern mesh

The medial construction method is then applied to the topological mesh. This method creates the 'in-triangle' of each mesh face by finding the mid-point of each edge and connecting these vertices and resulting in an 'open' polygon mesh (Fig.4).

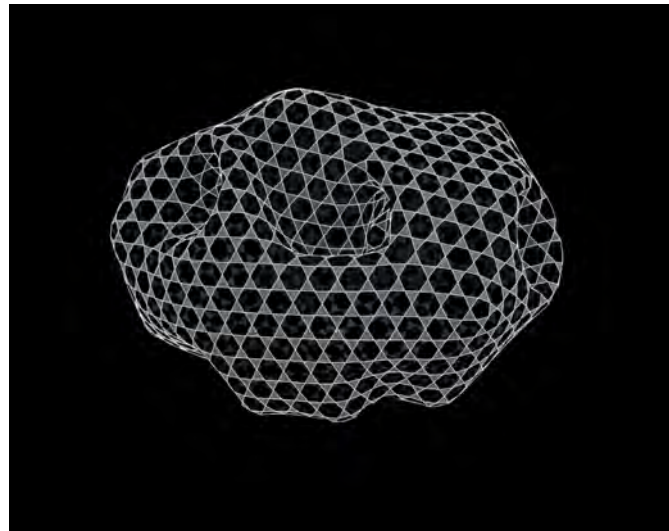


Figure 5: Kagome modelling, step 4: mesh relaxation with material parameters

The medial construction mesh is then relaxed using material stiffness properties as a constraint. This yields a geometric approximation of the weave design (Fig.5).

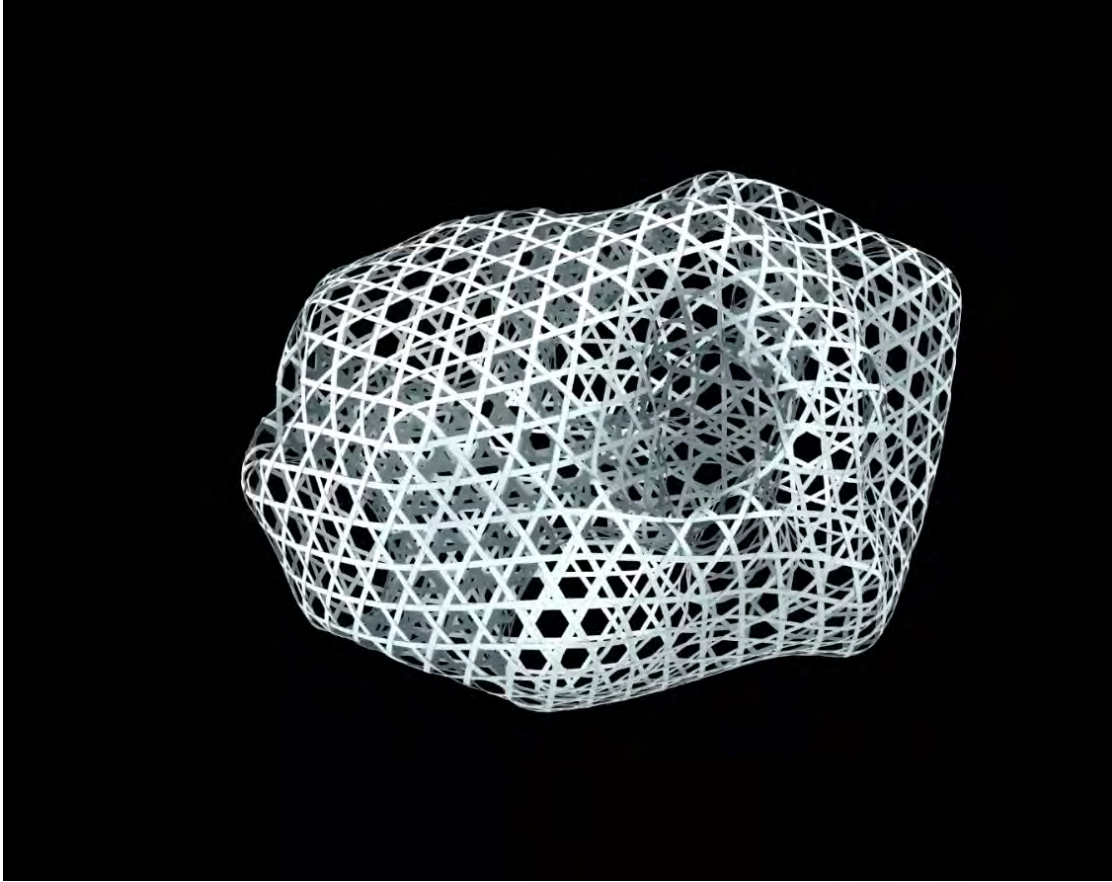


Figure 6: Kagome modelling, step 5: identification of weavers and modelling of interlacing

Identification of the weavers and their interlacing is easily accomplished with the 'open' polygon mesh. To accomplish this we exploit the 'right-hand-rule' that maintains edge direction for all mesh edges. We then elevate the start vertex of each edge along the vertex normal relative to the mesh surface. This is elevated by the interlacing height and then a new line is drawn between this elevated vertex and the unedited end vertex of the mesh edge. Performing a joining of all coincident line vertices results in the definition of independent weavers and derives an approximation of their interlacing. This saw-tooth geometry can be relaxed by generating a b-spline curve through the vertices, and, in turn, used as a guide spline to model the material dimensions (Fig.6).

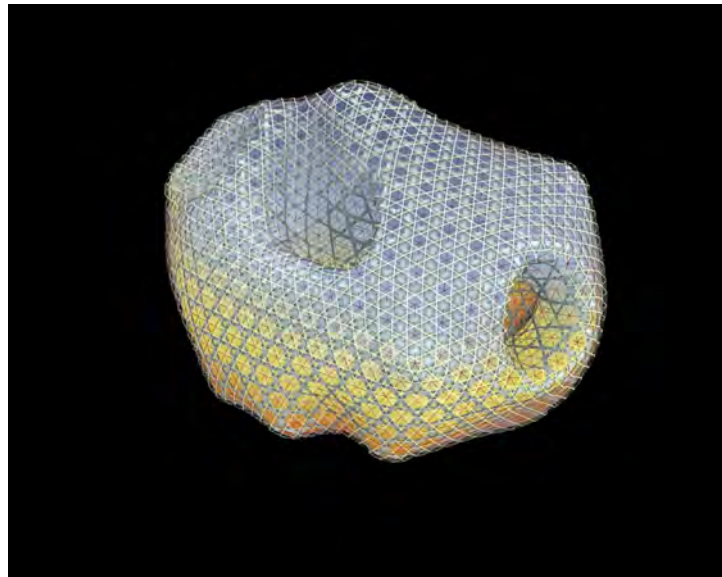


Figure 7: Kagome modelling, step 6: simulation driven mesh colouring

The Kagome weave is now fully modelled. Additional modelling steps can be applied to determine additional features, such as the positioning or specification of panels, for example. This can be generated using simulation approaches to colour the medial construction mesh (Fig.7).

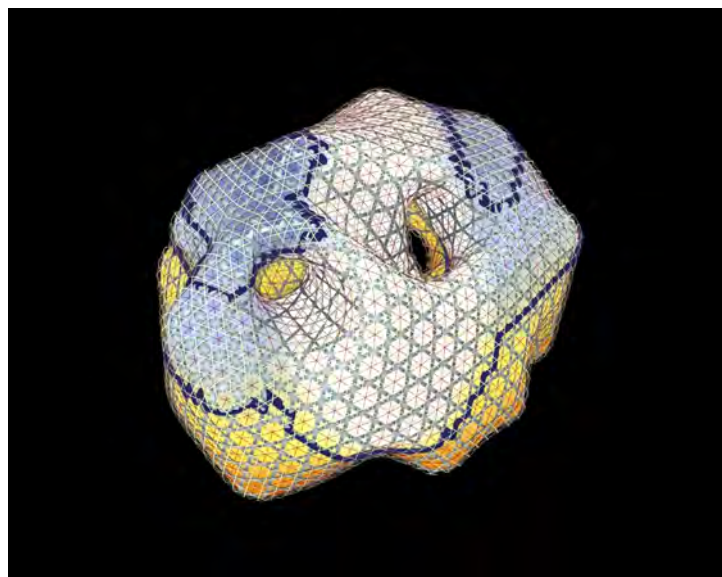


Figure 8: Kagome modelling, step 7: region discretisation of mesh colouring

A coloured mesh can be thresholded and discretised into regions for association with specifications according to design requirements (Fig.8).

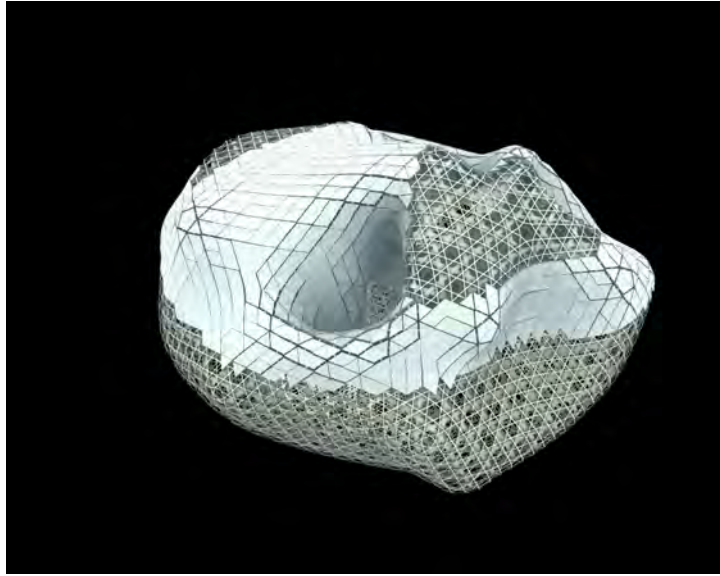


Figure 9: Kagome modelling, step 8: panelisation in desired regions

A descriptised region of weave is panelised (Fig.9).



Figure 10: Kagome modelling, step 9: sectioning to study internal

Sectioning through the model allows interrogation of internal relationships and qualities (Fig.10).

In Figure 11 we show various views of a fully modelled and principled weave pattern for an arbitrary high-genus morphology. The model embeds all necessary fabrication information and is parameterised to support further design investigation - for example the effect of different weaver sizes, or, more fundamentally, alterations to the weave topology.

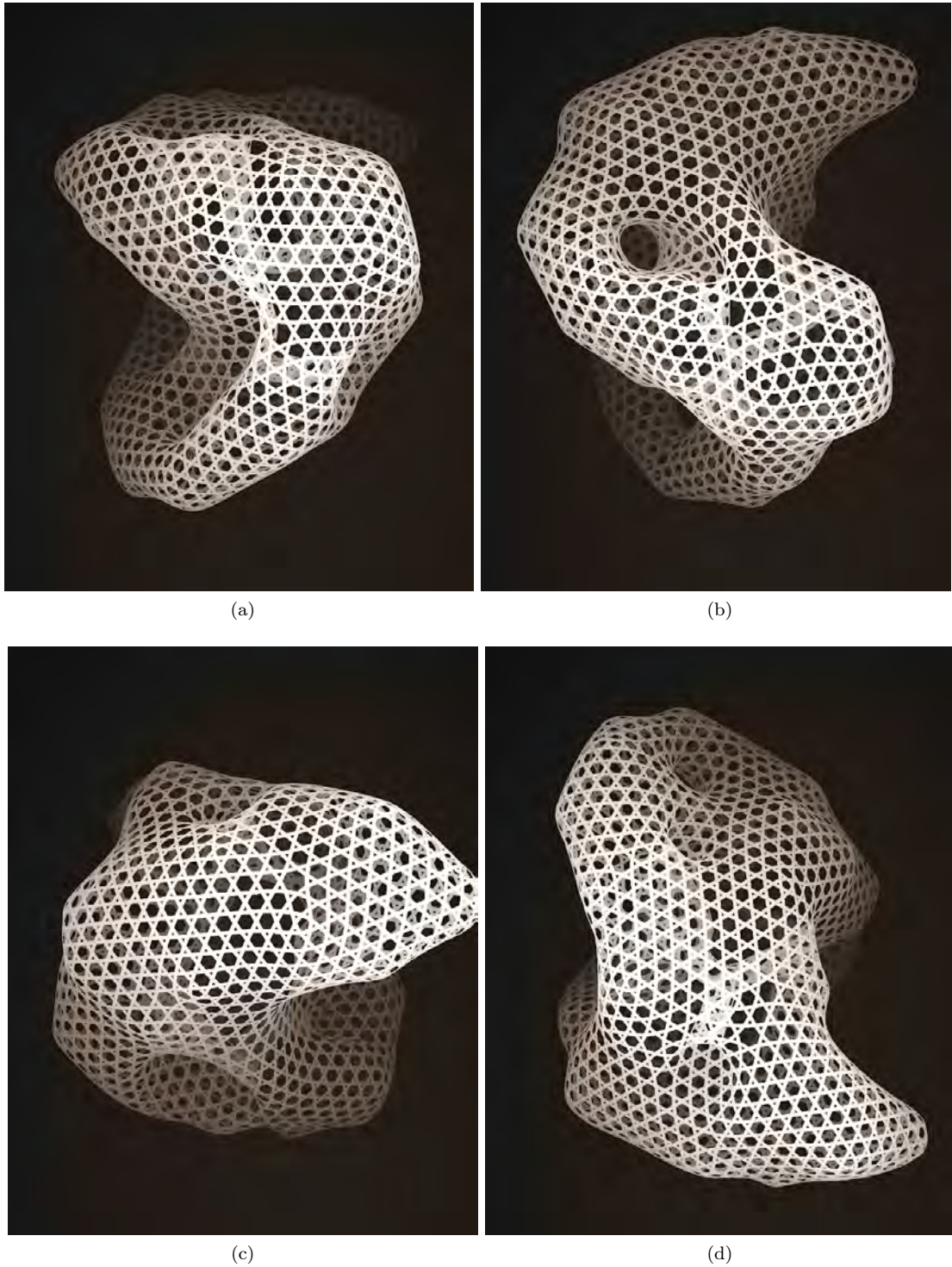


Figure 11: A principled weave pattern with weavers identified and interlaced for an arbitrary design target exhibiting complex geometry - various views (a-d). This model embeds all necessary information necessary for fabrication.

2.2 Limits in mechanical performance of MBC

In deliverable D5.2 - Report on preliminary characterisation and modelling, we reported on a material engineering review of the enzymatic activity of ligninolytic fungi [2]. Three principal material design strategies for MBC systems were identified: densification (by dense packing, cold or hot-pressing), composition (by introducing structuring elements, or modifying particles and/or fibre properties), and supplementation (targeting mycelium properties based on chemical tuning of the substrate). Of these three strategies, we conducted an experimental series targeting composition, which remains the least investigated strategy in the MBC literature. Drawing upon insights from the field of synthetic composite design, in which fibre composition is a primary design vector for functional property tuning, we reported on a substrate composition approach of orientated fibres in combination with different particle sizes and geometries for the bulk MBC volume (Fig.12).



Figure 12: Compression series. (left) bulk volume substrate particle sizes. (right) four post-compression samples illustrating different mechanical properties resulting from different structuring approaches.

We demonstrated the significance of introducing structuring fibres using approaches such as jacketing with hessian and internal reinforcement with reed or rattan fibres for both compressive and flexural behaviour. With an optimal substrate bulk material particle size of 0.75 – 3.0 mm, the introduction of common reed fibres coaxial to the load axis resulted in a 2.77-fold increase of the compressive modulus [3]. The flexural modulus of a longitudinally rattan-reinforced group of specimens was of 1.34 GPa for a density of 249.48 kg/m³ (control: 192.71 MPa, 232.24 kg/m³) [4]. It is anticipated that these preliminary results can be improved upon with hybrid material design strategies, for example, composition in combination with supplementation.

Despite these promising improvements in comparison to results reported on in the literature, the mechanical performance of MBC remains magnitudes of scale below that normally required for primary structure. Whilst we acknowledge below, in section 2.4, that design approaches exist demonstrating the exploitation of compression-only structural form to accommodate the poor mechanical properties of MBC in comparison to conventional materials used as primary structure for construction, our design criteria led to the decision of separating out the structural role from MBC materials and allocating this to a novel woven gridshell design.

2.3 Weaving with solid timber members

The ability to weave complex geometries using only straight strips of material, makes the Kagome method an attractive target for architectural application as it presents a rationalised and materially efficient production method that does not constrain the space of realisable morpho-

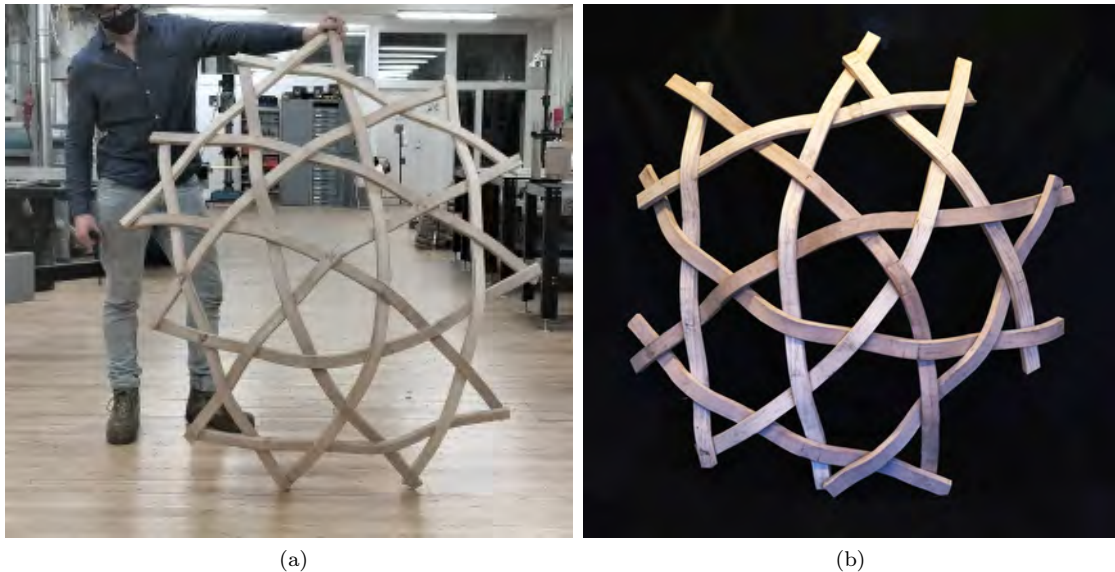


Figure 13: A valence 5 (note the pentagon, lower center) beech weave using timber members of 40 x 20mm cross-section (a) and and valence 5 weave produced with 60 x 30mm solid beech members (b). Note the plastic deformation of the members, achievable whilst they have a high moisture content.

gies. However, a key limiting factor to scaling up the method towards architectural application, without compromising the approach, has been the inability to find a material that is pliable enough to be interlaced, yet strong enough to perform within an architectural context.

Compwood - compressed wood - is a proprietary industrial technology for rendering solid timber extremely pliable. The timber is first heat-steamed and then compressed, under a large axial load. Once the load is released, the timber springs back to approximately 90% of its original length. Whilst the timber retains a high moisture content, it can be bent into extreme curvatures. The timber members have a maximum length of 3m, pre-compression, due to processing limits.

Solid beech and ash weavers of compressed timber with cross-section 40 x 20mm and 60 x 30mm respectively, were procured from the only supplier in Denmark.

Two weaves were produced using the 40 x 20mm timber - a valence 5 (yielding a synclastic surface) and a valence 7 (yielding an anti-clastic surface). A single valence 5 weave was produced using the 60 x 30mm timber. All weaves were assembled without the use of, or subsequent need, for mechanical or chemical fixings (Fig.13). This exploits the inherent jamming properties of the reciprocal triangles in the Kagome pattern, and the plastic deformation of the timber into sinusoidal geometry to realise material interlacing (Fig.14).

Due to the size of the prototype and the lack of suitable load-testing equipment to test it, an impromptu load-test was conducted without boundary restraints on the valence 5 weave composed of 60 x 30mm weavers. A load of c.250kg was applied around the weave's singularity and comfortably accommodated with minimal deflection noted.

Based on our empirical investigations, this approach to producing Kagome grodshells at architectural scale is very promising and novel within the industry. It also directly translates the Kagome method to larger scales without compromising aspects of the approach, making craft skill directly transferable to larger structures. This embeds the core values set out in the



Figure 14: Detail of the interlacing on the valence 5 weave using 60x30mm solid beech members. Note the plastic deformation of the members, achievable whilst they have a high moisture content.

introduction of developing an inclusive, equitable, affordable, sustainable construction approach with low barriers to entry and built upon principles of ecological and social justice, community practice and empowerment through accessible craft mastery.

Further research is required to scientifically determine the mechanical performance of weaves at this scale. Additional areas of focus should target long-term performance and degradation of the weaver material as no data exists for this timber treatment method applied to weaving at this scale, as far as we are aware.



Figure 15: An impromptu load test of c.250kg applied to the singularity of the weave made using 60x30mm members. No boundary restraints were added for the test. Minimal deflection was experienced.

2.4 Speculative design investigation for a Fungal Architecture

As reported in deliverable D2.3 section 3 (D2.3 - Structural soundness is verified), the literature provides evidence of design methods for utilising MBC in load-bearing roles for structurally informed spatial configurations. However, a key limitation is that structural form must be constrained to compression only designs [5, 6, 7], restricting the space of possible morphologies that can be expressed.

For the FUNGAR project, this represents a fundamental restriction in the possible deployment of computationally active materials. A key finding of the FUNGAR project, by partner UWE, has been that living mycelium-based materials exhibit different spiking characteristics based on stimulation with light and mechanical loading respectively [8]. The consideration of this finding in the context of architectural application, suggests that placement of living mycelium in specifically designed locations could lead to qualitatively differentiated patterns of activity resulting from a given morphology in interaction with its environment. This aligns conceptually with theoretical understandings developed within the field of Morphological Computation [9, 10, 11, 12] in which morphology is understood as computational resource which can be designed with the aim of ‘out-sourcing’ computational control requirements. Therefore, we have developed our construction logic to permit the largest possible morphological design space, with the assumption being that responsive programming for a living architecture can be partially approached through the consideration of a design form within a given geographical and embodied context.

Initial explorations were conducted to search for morphologies that could give rise to specific information generation characteristics, given a specific geographical location and assuming sunlit conditions. In figure 16 a simulation is conducted to search for positions to place living mycelium that can be triggered through light stimuli to produce a signal. A number of points are sought to embed a notional ‘majority-rule’ logic. In figure 17 the positioning search space is restricted to a particular spatial region. These initial studies provided a conceptual basis for considering how living mycelium, with its sensing capabilities, could be used as a design driver and impact macro and meso-scale geometries for a Fungal Architecture. In combination with the Kagome weaving, which provides a rationalised and materially efficient approach to realising approximations to any complex manifold morphology, we have the basis for developing a unique architectural expression and construction concept that embeds novel functionalities based on novel materials.

From this basis, we conducted an iterative design exploration to refine the construction concept and its expression through use-inspired scenarios. As a first step, we explored how the conceptual separation between load-bearing structure and enclosing skin could be made spatially manifest. We accomplish this by considering the roles of structure and enclosure being served by two different Kagome weaves (Fig.18). By interlacing these weaves, we create the opportunity for a rich and varied topography to be established. Selective colonisation of weave for enclosure allows for light and view to be programmed into the surface (Fig.??).

Figure 19 shows a detailed study of the relationship between the primary gridshell Kagome and surface enclosure performed by a mycelium composite layer sandwiched between two Kagome weaves.

A notional site was chosen to exercise these design rules in the design of a hypothetical architectural proposition. The proposition develops a strong landscape quality with an undulating surface interacting with the ground condition and creating a variety of elevated, sunken or surrounded conditions. The spatial organisation is guided by principles of asymmetric and wandering routes leading to contained spaces. Internal spatial divisions are determined through single curved walls that exist independently from the Kagome gridshell structure. This provides planning flexibility and differentiation from the geometry of the structure above.

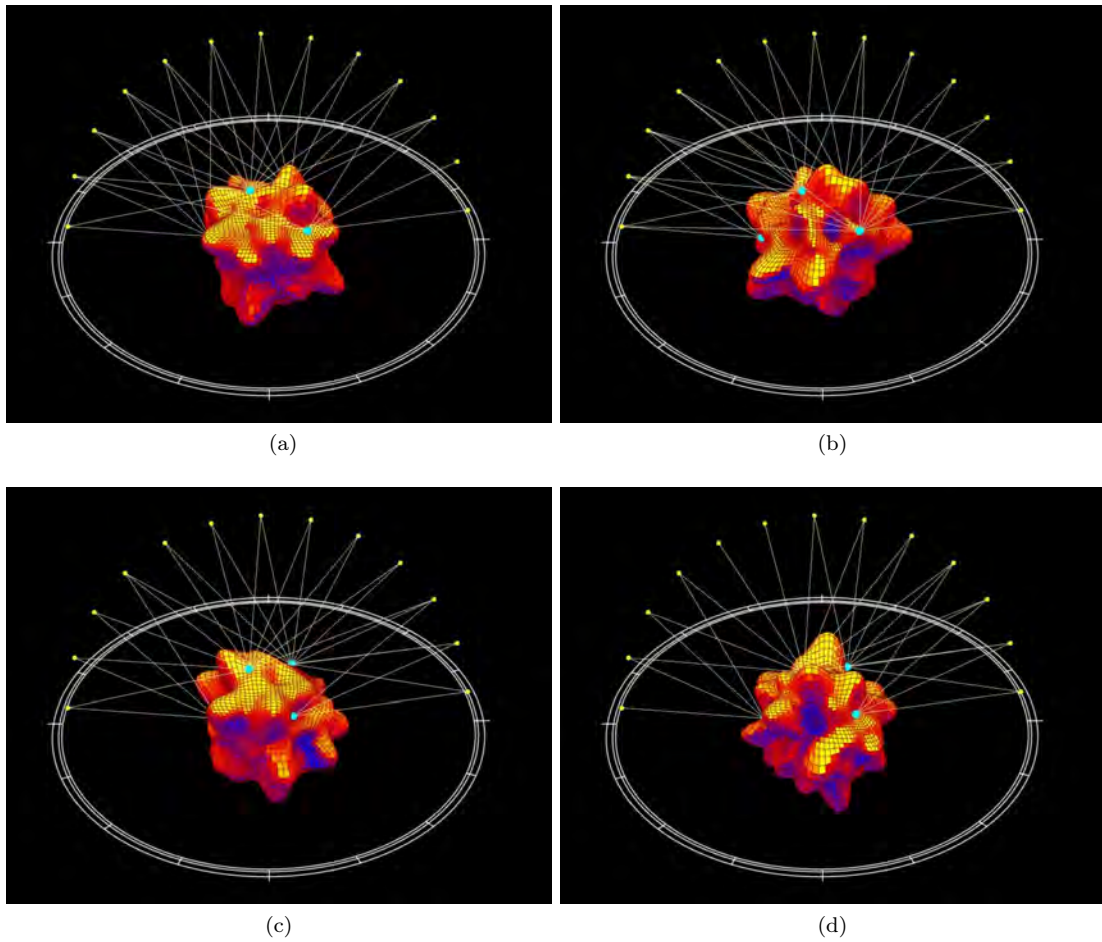


Figure 16: Searching a morphological design space for placement locations of living mycelium given a specific geographical location.

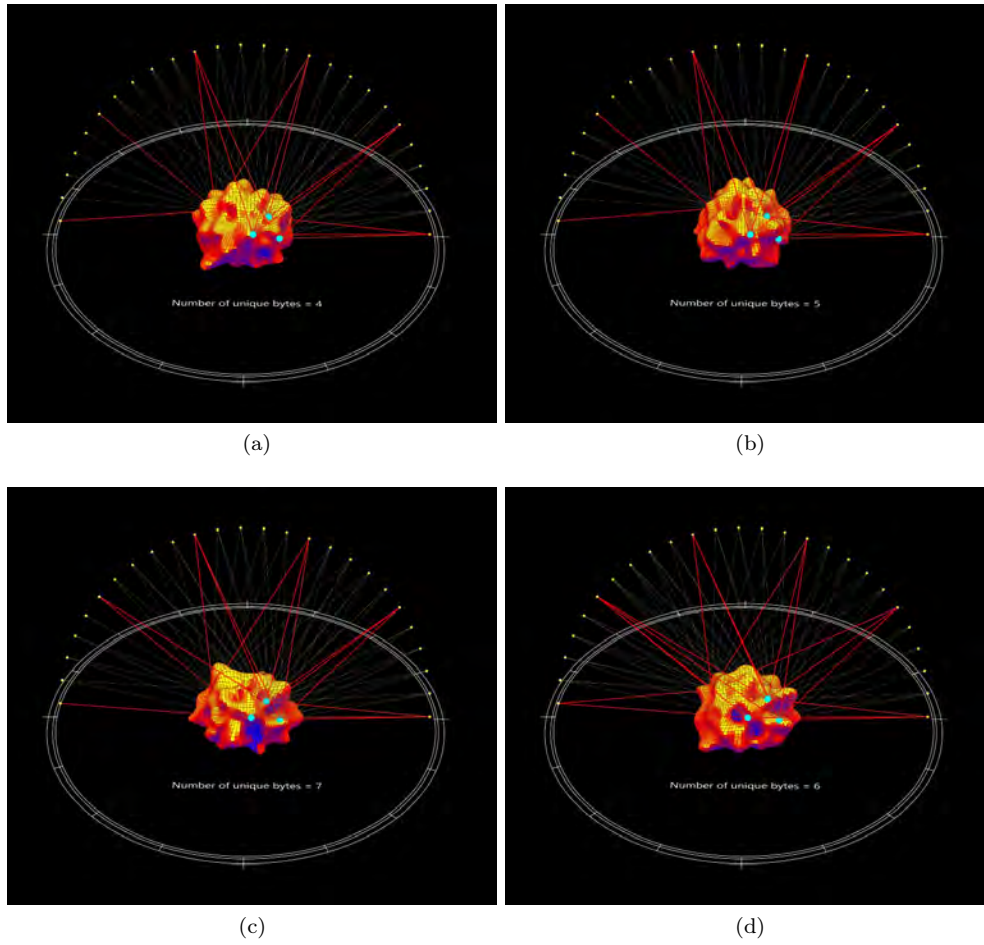


Figure 17: Searching a more constrained volume of space for the placement of living mycelium locations.

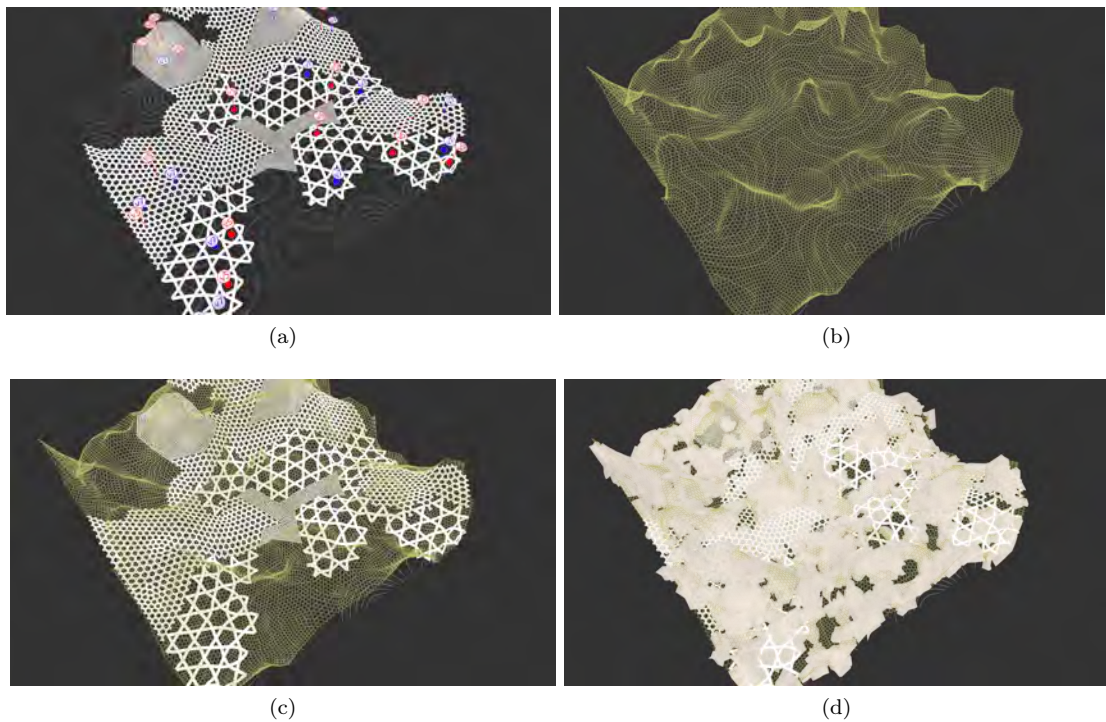


Figure 18: Revised construction concept with a primary structural support provided by a Kagome gridshell (a), a secondary weave that to support a mycelium skin (b), the interlacing of the two weaves to create varied spatial conditions (c) and the selective population of the secondary weave with mycelium (d).

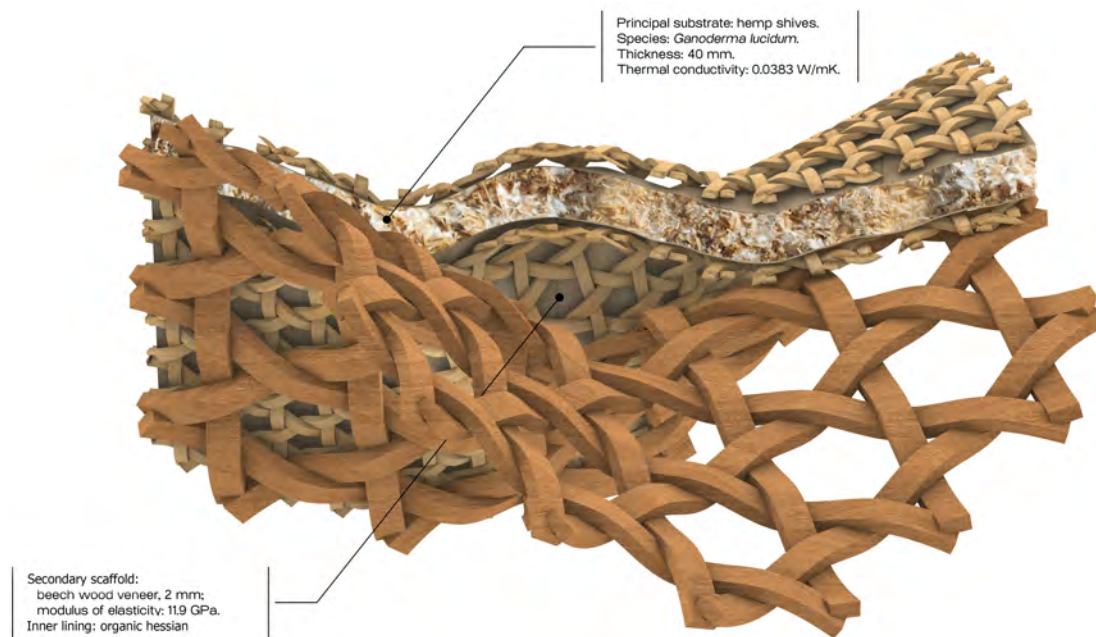


Figure 19: Detail study showing the relationship between primary structure and building envelope.

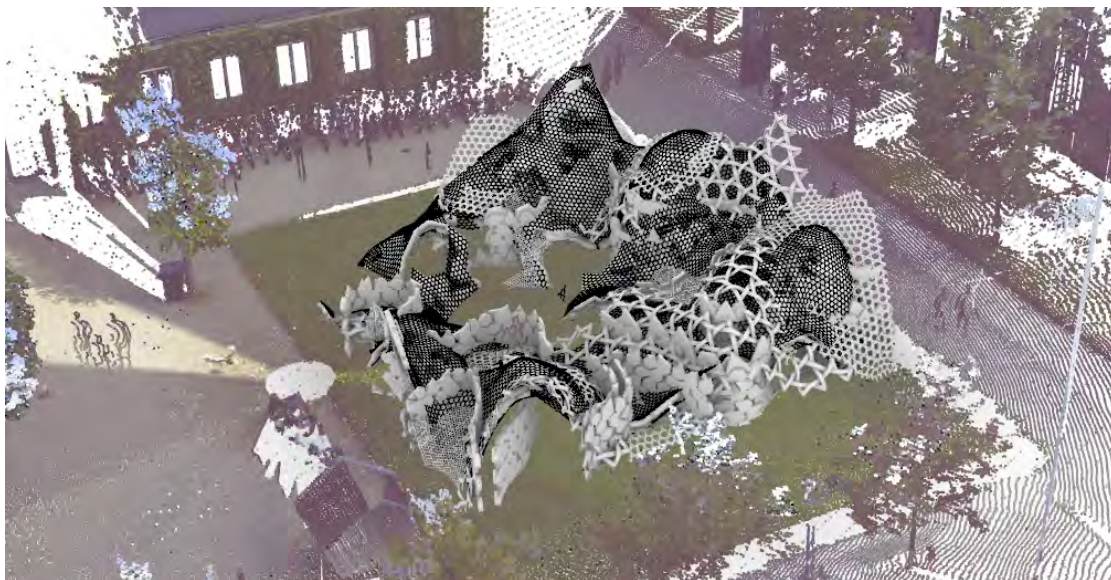


Figure 20: Perspective view of the speculative architectural proposition.

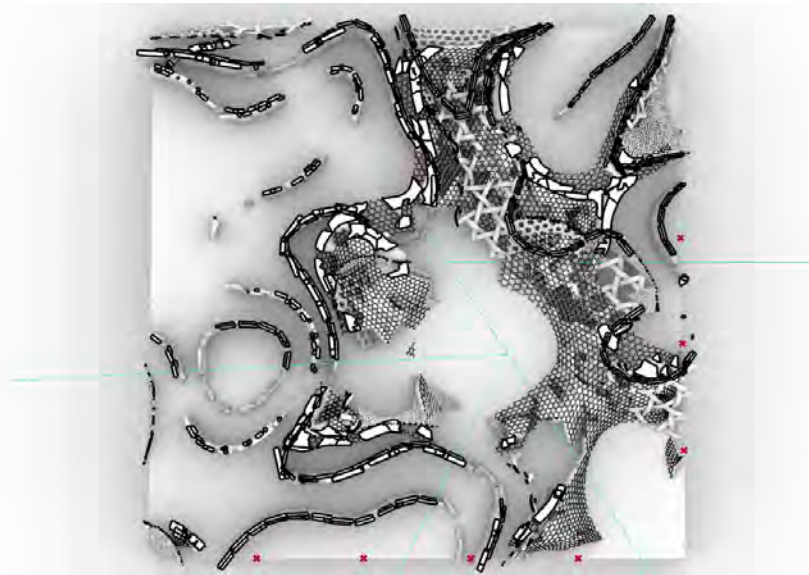


Figure 21: Plan view of the speculative architectural proposition.

2.5 Conclusions

In this section we have reported on key aspects of design investigation that have developed and revised our original construction concept. Our revised Kagome modelling method provides a generalised approach for generating principled Kagome weave patterns to achieve any morphology that can be modelled as a manifold polygonal mesh. We have determined that MBC composite mechanical performance is not adequate for use as primary structural material in our case. We have demonstrated a novel Kagome weaving approach to realising timber gridshells using solid timber weavers, that preserves the principles of Kagome production. Finally, we have integrated these findings to develop novel spatial design vocabulary for Fungal Architectures.

This work establishes the basis for the design rules which have been encoded into a digital design workflow described in the following section.

3 DESIGN RULES

3.1 Workflow

In this section we describe the integration of design rules into a computational workflow for designing Fungal Architectures. The workflow is built around the consideration of four areas of design influence:

1. structural analysis
2. spatial division
3. environmental analysis
4. identifying sites for living mycelium acting as sensors and signal generators

In the following subsections, we briefly describe their operation and dependencies.

3.1.1. Structurally informed design

A spatial envelope is created by moving the control points of a flat surface vertically, resulting in a series of random and structurally inefficient forms. A maximum cantilever distance and angle is prescribed. The most efficient point outside the supported region is identified (by proximity to the floor and to other surrounding features) and moved to the floor, creating a structural column that can incrementally increase in radius as the surrounding control points descend until the span is within the maximum bounds. If descent exceeds a maximum iteration count, the strategy is disregarded and new points are selected. If this fails repeatedly the iteration is ignored. It also considers a constraint for minimum heights for habitation. Once the surface is defined, a primary triangulation occurs and is relaxed to achieve a morphology achievable with Kagome. This is then tested structurally analysed to determine viability. Modifications are made to the control points and the process repeats. Once a structurally viable form has been established, the secondary weave is generated through an interference pattern which creates a surface that intersects the primary structure, with a minimum frequency and maximum amplitude ensuring the distance directly between support points, and geodesically across the secondary weave, do not exceed a predetermined maximum. When the secondary weave is derived from the primary mesh it is possible to ensure the weavers do not intersect and so the two layers are topologically independent but sparsely interwoven. Figure 22 illustrates the steps and potential feedbacks for this component of the design workflow.

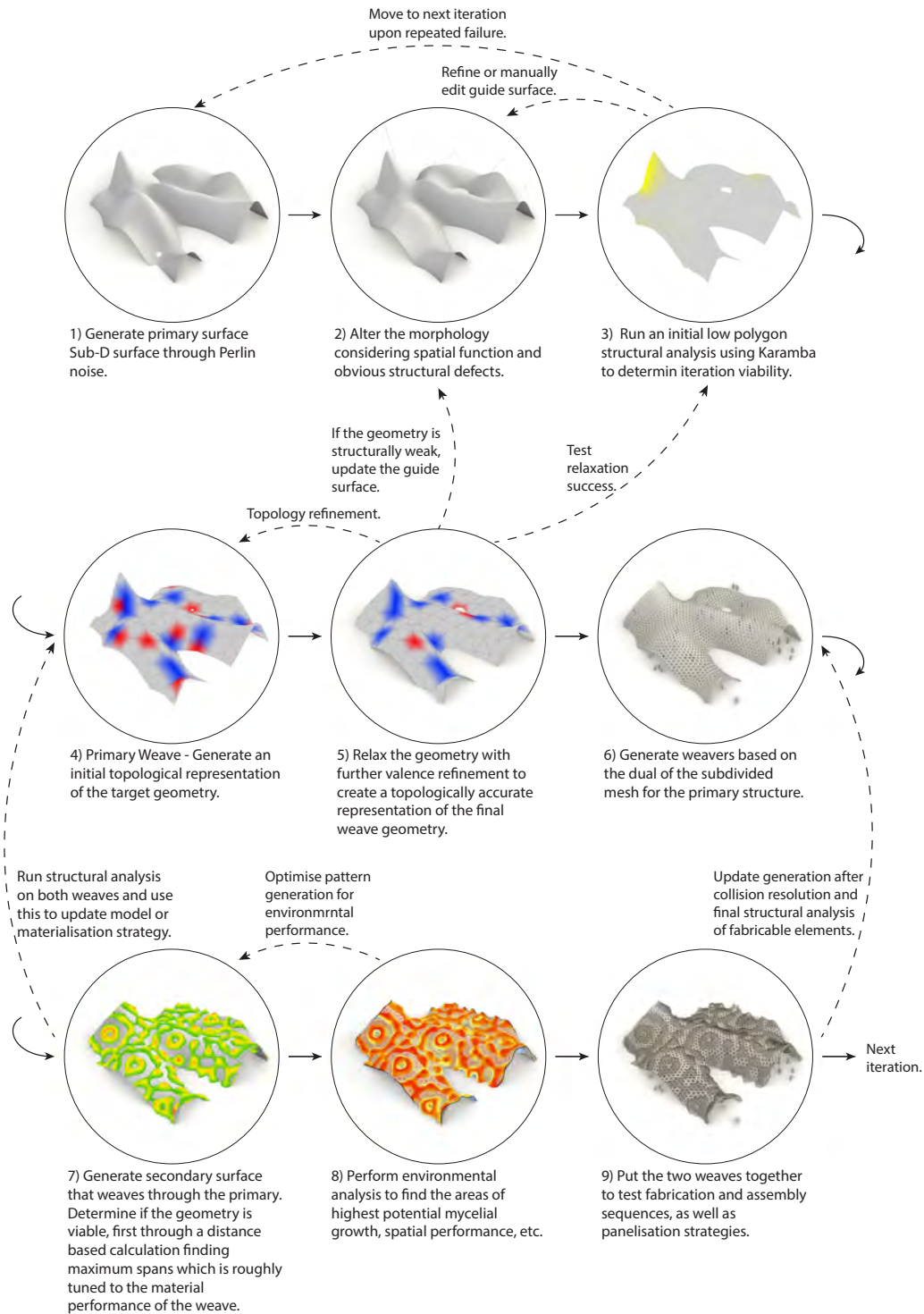


Figure 22: Kagome structural modelling workflow

3.1.2. Spatial intentions

Having adopted a Kagome gridshell typology as the principal spatial and structural concept, a key consequence of this is the need to consider a strategy for the subdivision of the spaces defined by the gridshell. Here, we make use of a moulded panel design (see fig.23) that can be tessellated (see fig.24) and operate independently from the gridshell (see fig.25).

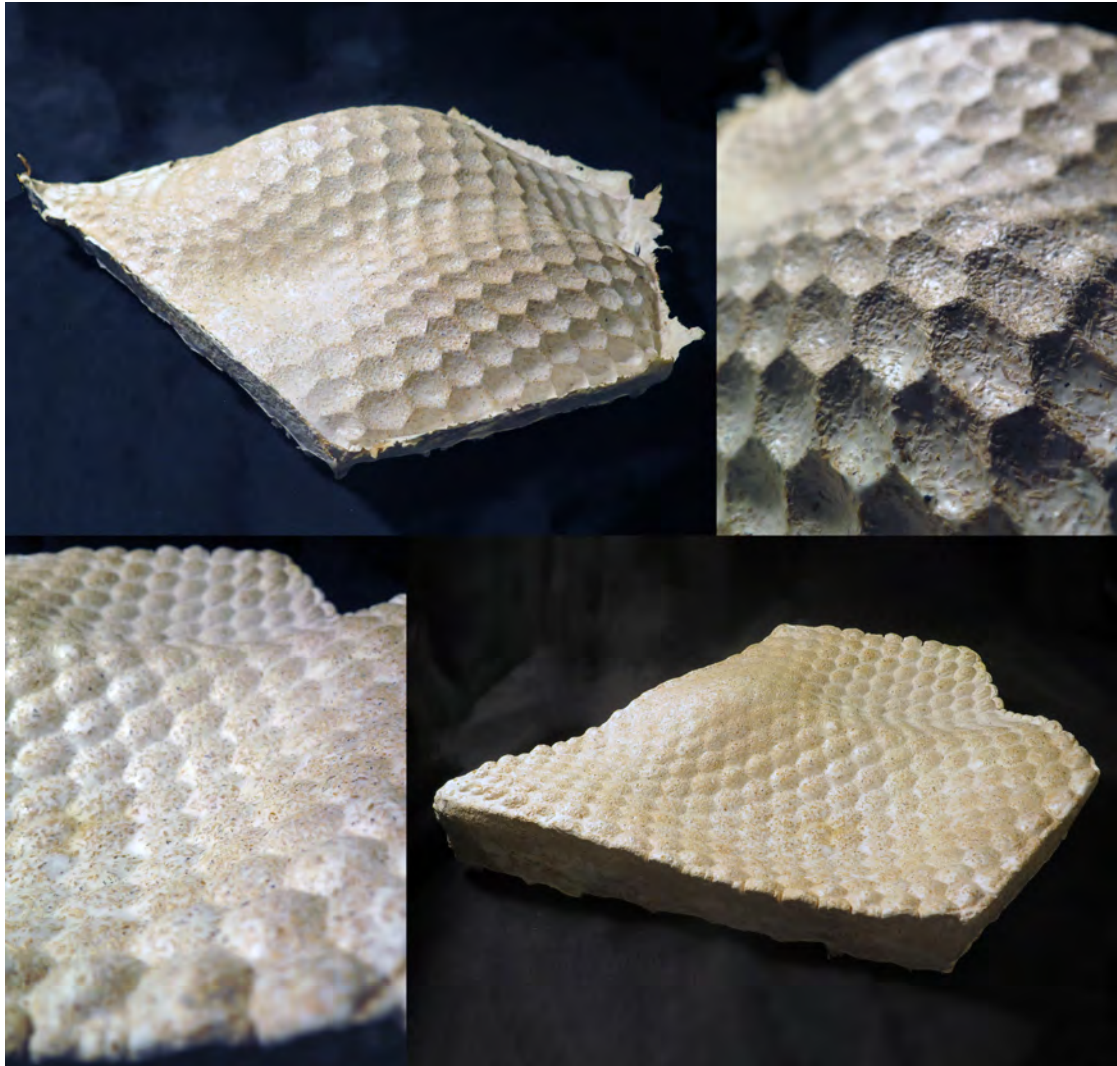


Figure 23: A concave and convex panel design with insets showing surface detailing. The panels can be tessellated to create internal subdivisions.

In this way we can define walls and subdivisions arbitrarily. These spatial dividers can only exhibit single curvature. Their geometry is defined by a polyline that drives a surface tessellation algorithm.

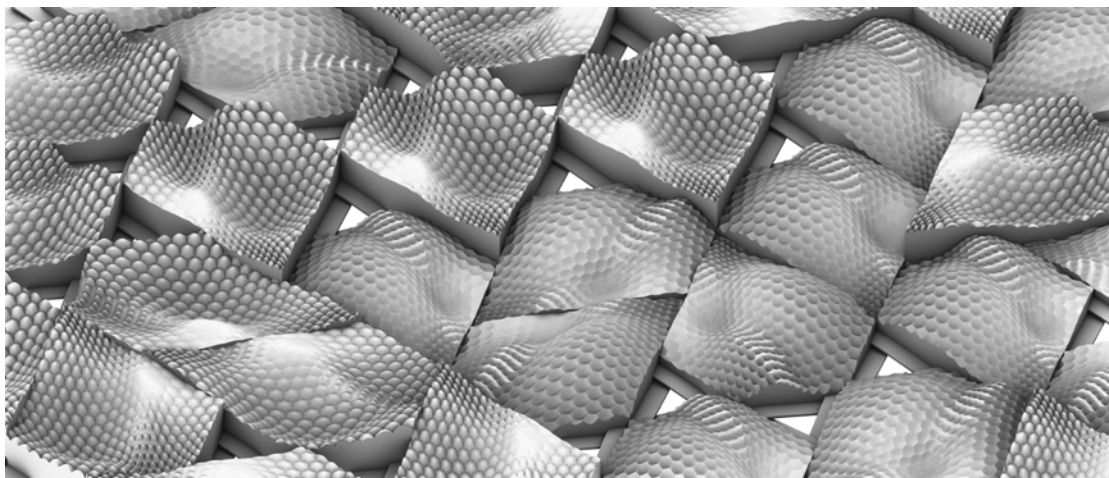


Figure 24: Rendering of a planar tessellation of panels with convex and concave distribution. Their design consciously avoids trying to make a continuous surface in the tessellation, providing a contrast to the geometry of the Kagome weave.

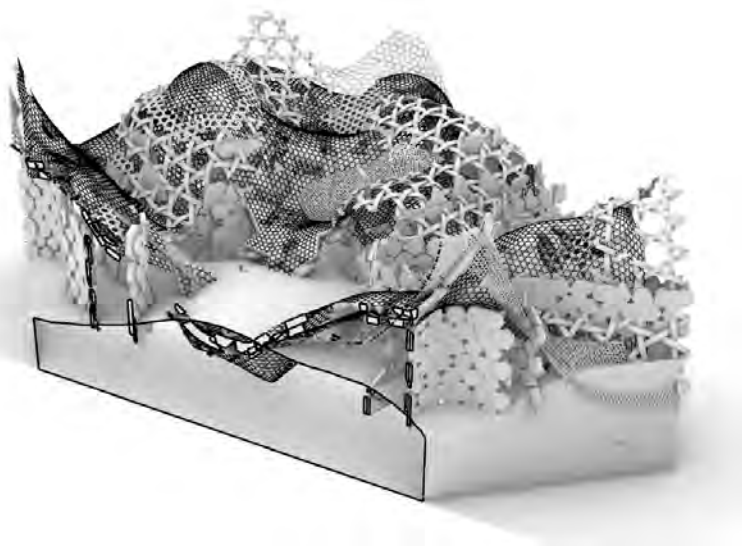


Figure 25: Subdividing walls made of tessellating panels are used to sub-divide the space of the gridshell.

3.1.3. Environmental analysis

Given a particular geographical location, the predominant prevailing wind directions are can be constructed from historical data which can be further refined to include the influence of the immediate surroundings. These are then imposed on the form which responds with deformation to amplify the negative pressure created on the leeward face. This could increasing the draw of strategically placed vents and passively cool the interior environment. These ridges are also useful in the external collection and storage of rain water, making a suitable environment for further colonisation of the external surface by other flora. In the future, this could also be a possible trigger for a behavioural response of the building, or extending the analysis to other environmental variables. The current structural generation guides the designer towards viable morphologies and excludes those that are not viable. It is however a holistic process for generating a primary and secondary weave that responds to environmental stimuli and constraints, and is an apt demonstrator of the proposed system. In the following section, we extend the environmental analysis to help inform the placement of placement of components, specifically living mycelium for active sensing and signalling.

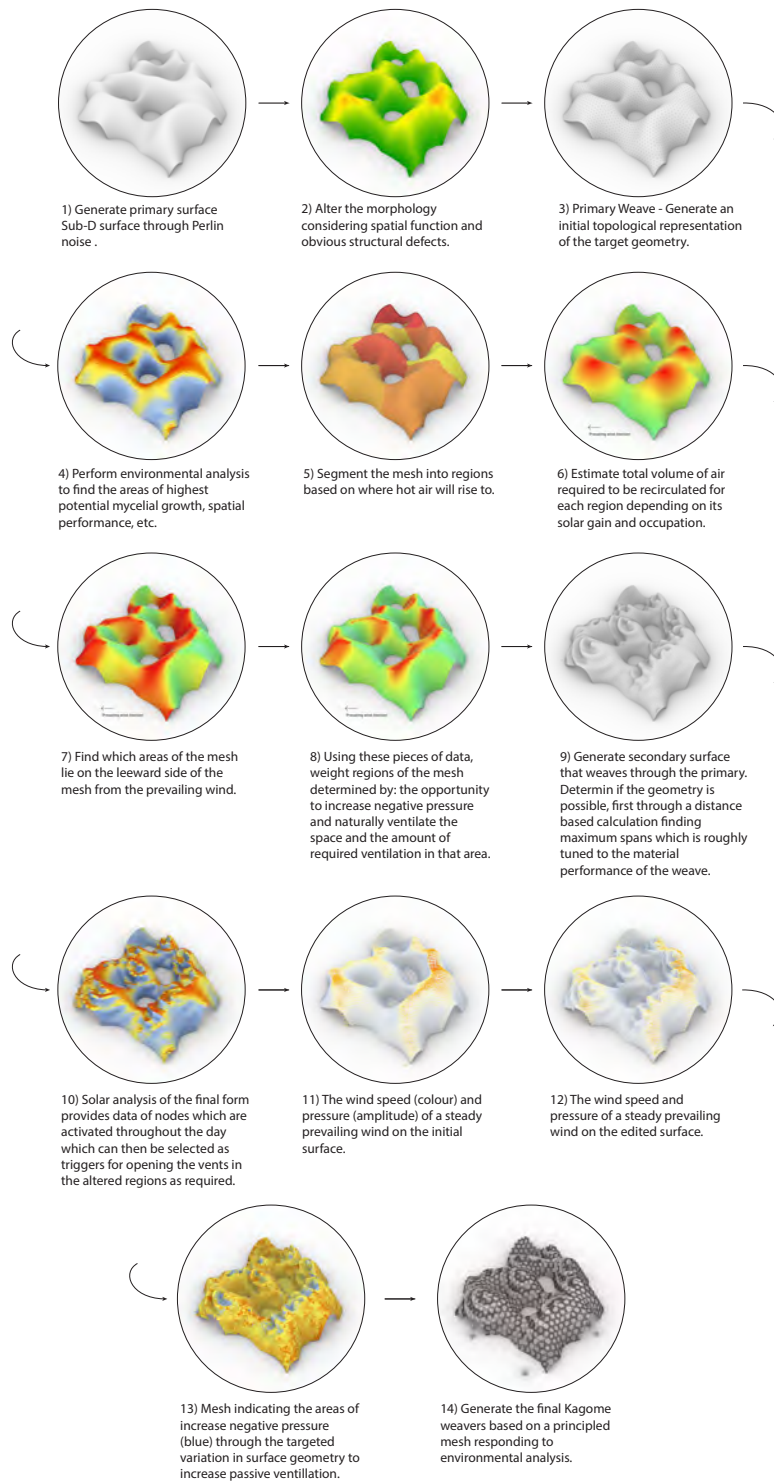


Figure 26: Kagome environmental modelling workflow

3.1.4. Identifying fungal sensing sites

In this portion of the workflow, we embed the possibility of determining sites for placing living mycelium in conditions conducive to generating environmentally driven signals, but also creating local morphological modifications and, hypothetically, acting as an environmental control for passive internal cooling.

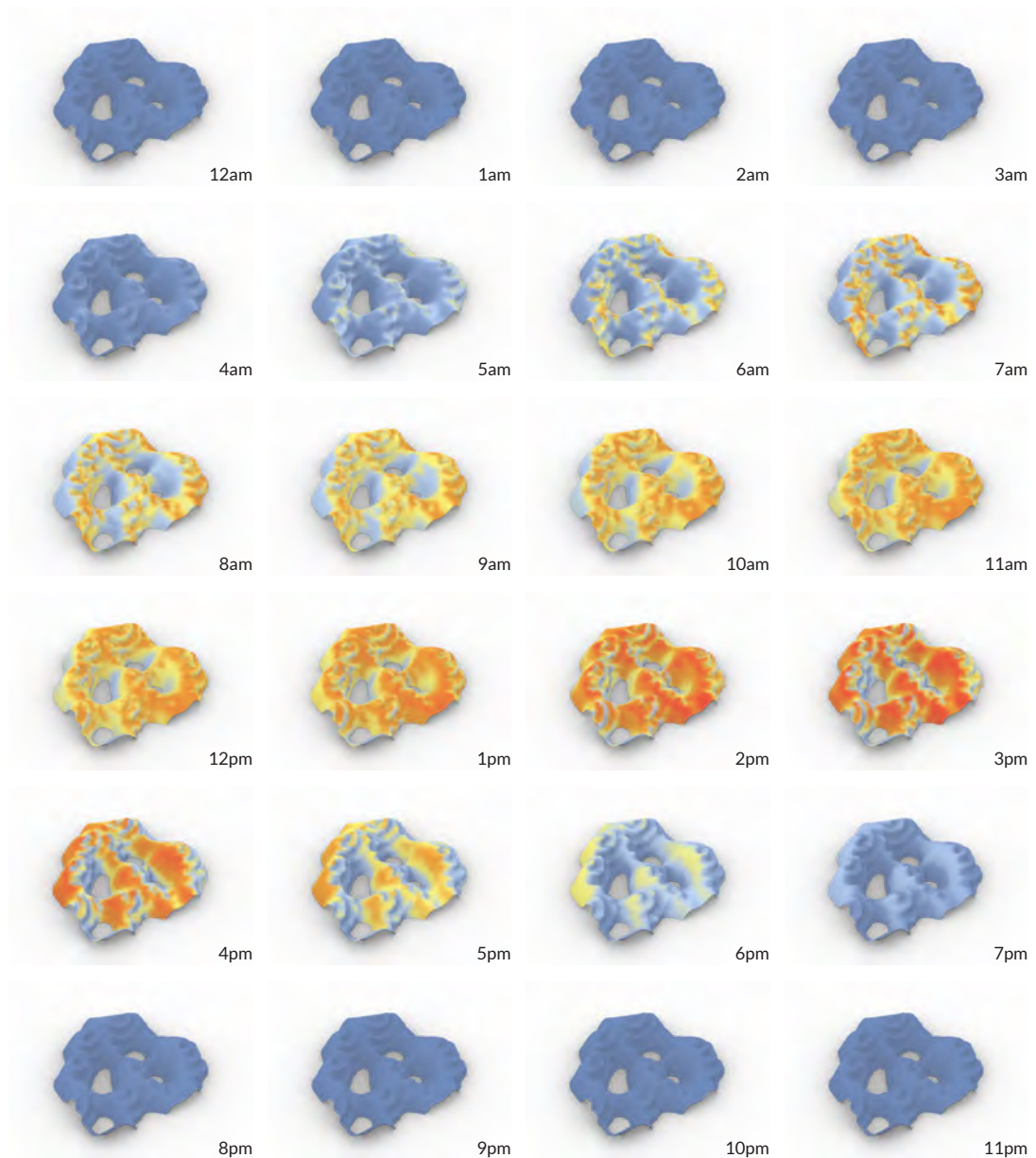


Figure 27: Total radiance for each hour of the day in July

Solar radiation is calculated through the day on the mesh from which the secondary weave is

generated. The surface articulation provides rich variation in surface activation throughout the day, with regions periodically exposed to sunshine and shade as the sun moves. The cumulative heat is accounted for with regions exposed to solar gain without shaded periods showing greater activation, and with regions at the peak of the surfaces collecting hotter air as it rises inside the structure, which indicates locations for potential vents to disperse this heat.

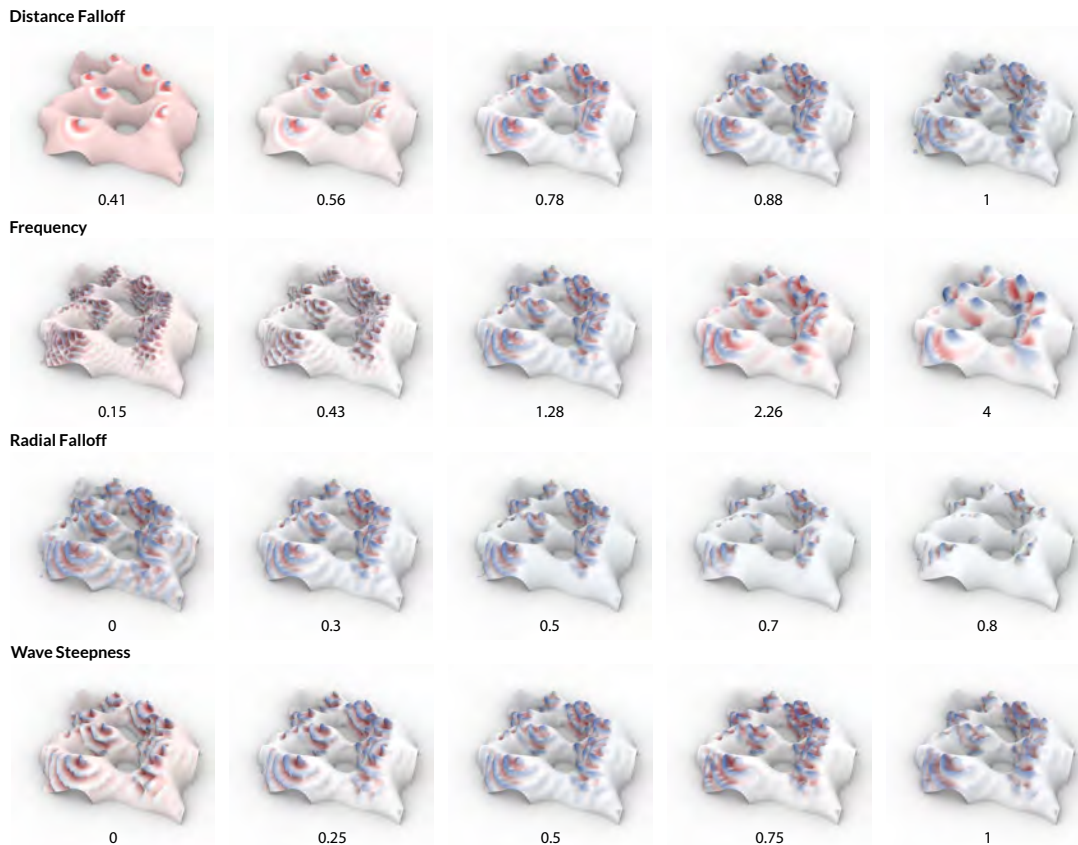


Figure 28: Wind driven surface articulation

All nodes are tested to find their solar activation through the day. For each hour, each possible combination of the different combined states are calculated. From this the nodes are selected that between them provide the largest number of different state permutations with the minimum number of nodes throughout the day. In this simulation, activation is binary. However there is the possibility to account for a spectrum of solar activation states, as well as the cumulative impact through the day on substrate moisture content and signal variance, as well as responding to other environmental stimuli.

3.2 Integration of sensing-lighting system

In this investigation we leverage the Fungal bio-electrical sensing methodology of partner UWE towards architectural application. We identify a key design challenge being how to communicate



Figure 29: Environmentally driven node selection

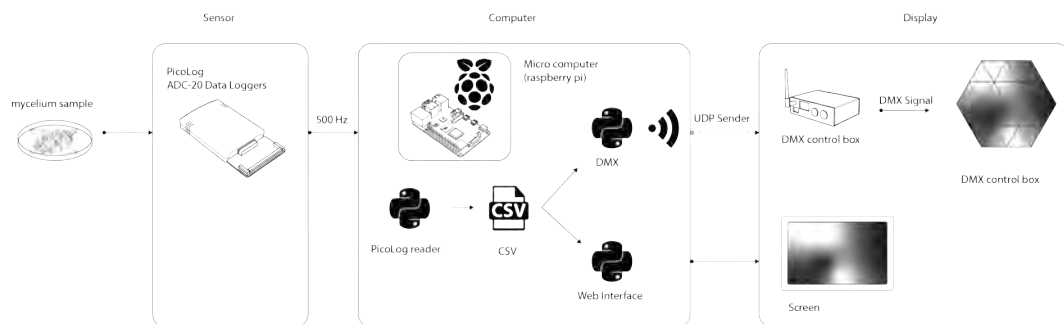


Figure 30: Sensing-lighting workflow

the living nature of mycelium to occupants of the architecture. The bio-electric activity of mycelium provides a near real-time adaptive signal that could be used to drive ephemeral spatial effects. We achieve this by measuring the electrical signals generated by mycelium and representing these through a web interface and lighting device (Fig.30). The lighting device is based on a proprietary and patented system developed by artist US Jason Krugman, with control hardware developed by Matt Mets of Blinkinlabs.

3.2.1. System specification

To measure the mycelium activity, a high-resolution data logger equipped with a 20-bit A/D converter and custom scripts is used, along with a pair of stainless steel needle electrodes. The electrodes are inserted approximately 15 mm to 2 cm into the substrate. The activity is sampled at a rate of one measurement per second, and the average value is recorded. The measured values obtained from PicoLog are saved in a CSV file format on the Raspberry Pi. Typically, the sampling frequency is set to 500 Hz, and the obtained values are mapped to an RGB color gradient (see Fig.31).

The lighting device utilises the Digital Multiplex (DMX) protocol, which is a unidirectional lighting control protocol enabling sophisticated lighting sequences to be controlled from a single source. The control box (Supersweet, Blinkinlabs), is equipped with a built-in WiFi connection and provides ten outputs, each referred to as a universe. Each universe can accommodate up to 512 channels, with three channels dedicated to controlling the RGB LED strip.

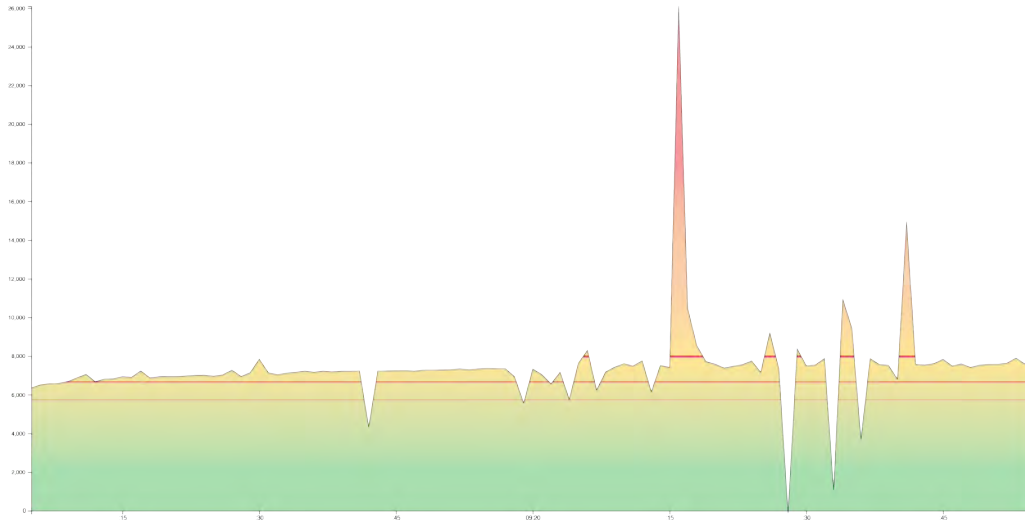


Figure 31: Sensing data and responding colour gradient

3.2.2. Mapping of sense data to output

Regarding the mapping of sensory data to output, simply mapping the activity value within a fixed range may result in low contrast lighting, where the colour representation only shows spikes or remains the same throughout (Fig.32). To address this, we employed a data windowing technique, or 'bucketing', to record values over a period of 1 to 3 minutes. This allows us to determine the average, highest and lowest values, which are then used to adjust the color gradient (Fig.33). The lighting animation typically operates at a frame rate of 30 to 50 frames per second. The signal values received from Picolog are stored in an animation CSV file, resulting in a delay of approximately three minutes in streaming the activity to the display, due to the signal collection required for the animation.

The display presents the activity from the past three minutes, and the frame rate of the animation is controlled based on the current average values, whether they are decreasing or increasing. The rhythm rate ranges from 23 to 40 frames per second. The web interface integrates both the electrical signal information from the mycelium and environmental data. The background of the interface visually represents the current activity, while the palette in the bottom-left corner provides the user with various color options. The past activity records are stored in the lower left section, allowing users to click and playback previous activity.

3.2.3. Temporal progression of patterning

The various patterns showcased in figure 35 exhibit distinct ambient animations. Options A, B, and C employ the same lighting logic, where the gradient changes correspond to the transmitted reading signals.

The motion vector illustrates the current activity of the mycelium sample, indicating whether it is high or low. In Pattern D, the right side of the display showcases past moments of activity, while the left side represents the current reading.

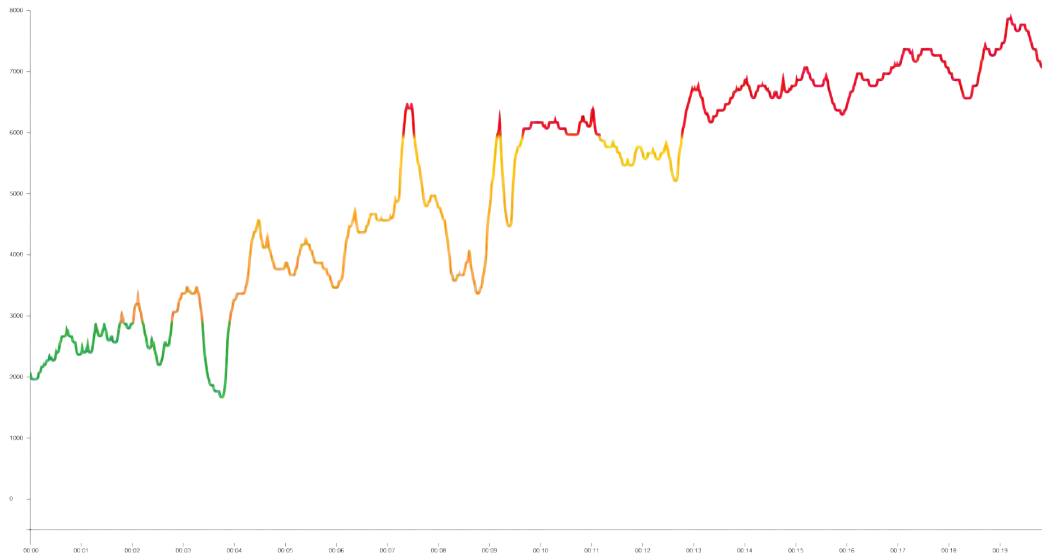


Figure 32: The colours with the same threshold in ten hour period of data.

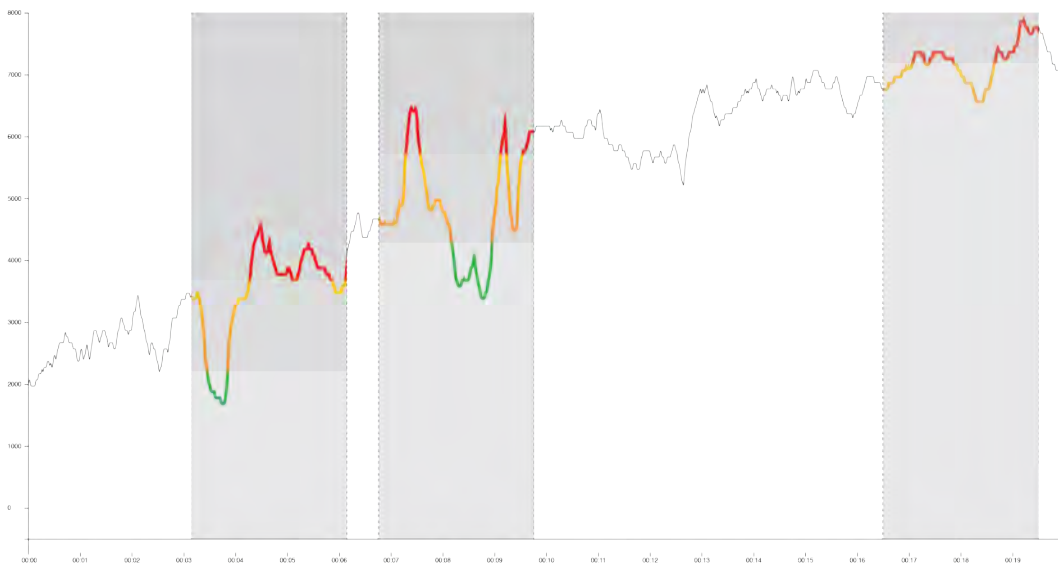


Figure 33: Bucketised data window and the resulting threshold adjustment to the colour gradient.

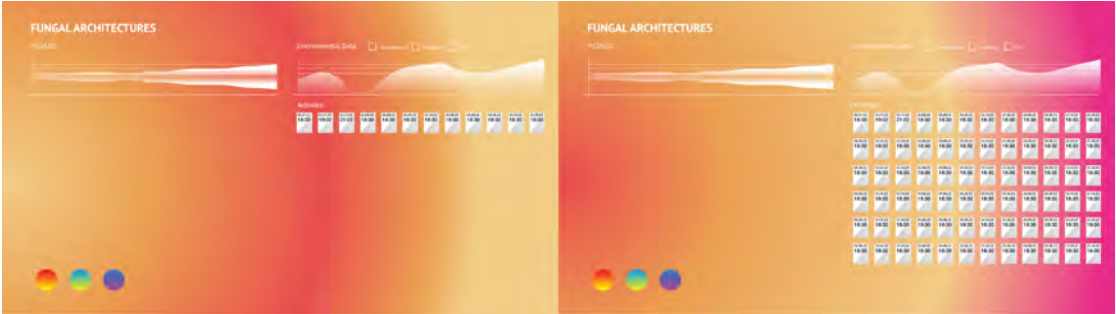


Figure 34: Web Interface of mycelium activity.

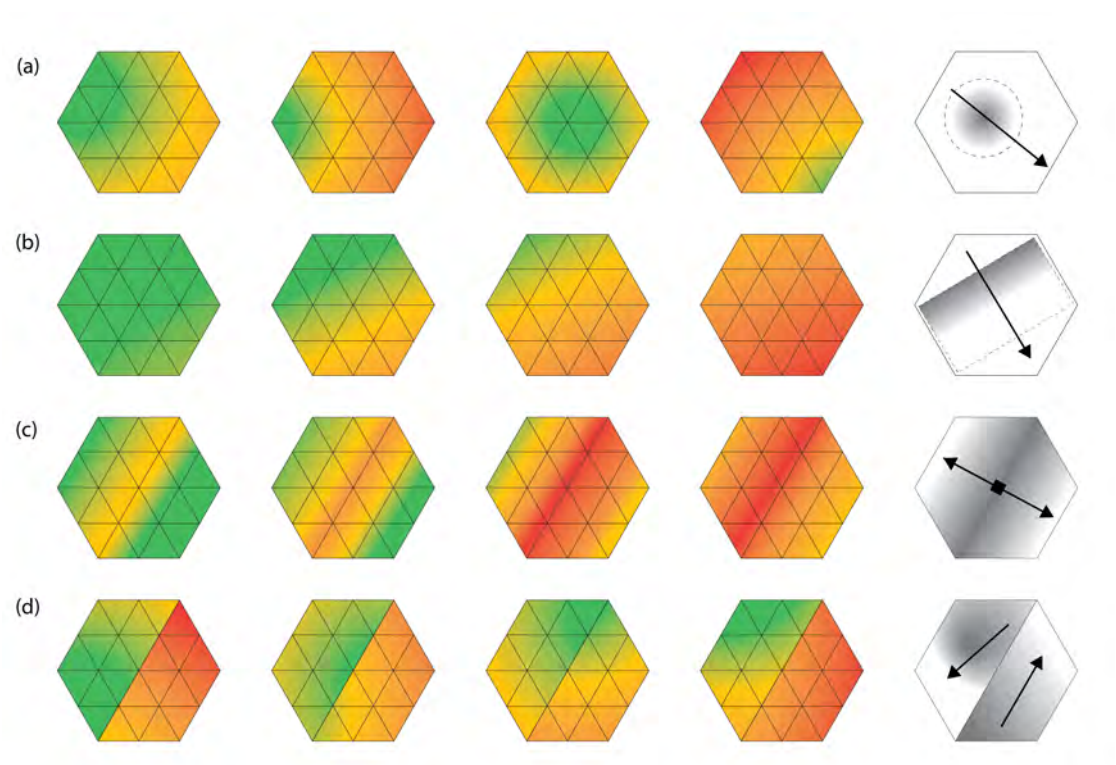


Figure 35: Four studies of different lighting patterns generated by differential mapping of mycelium bio-electric signals into the lighting network.

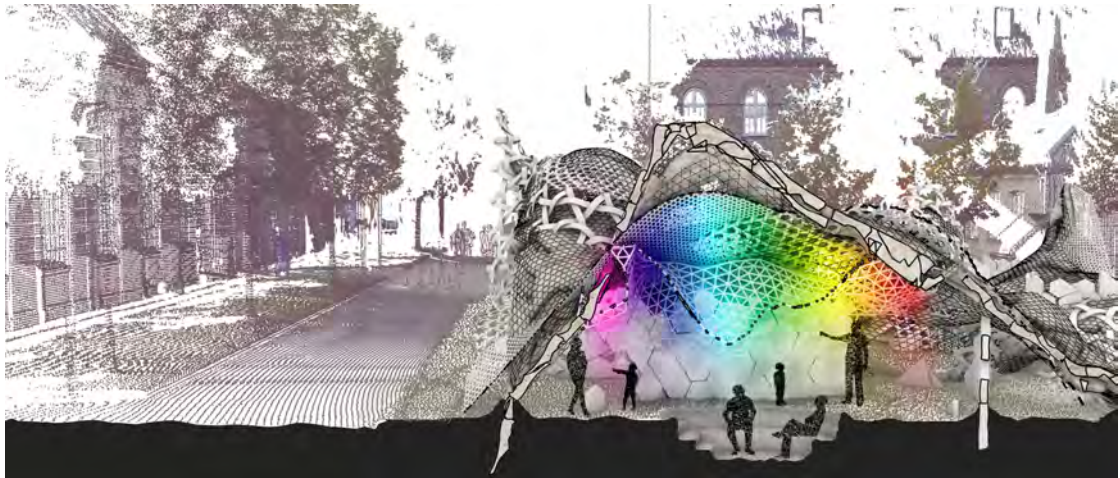


Figure 36: Speculative rendering of a 'conversation pit' within a Fungal Architecture, with the lighting system acting as a conversational partner.

In figure 36 we illustrate how the lighting hardware could be incorporated into a Fungal Architecture. The lighting system is placed above a 'conversation pit' - a prominent architectural feature in mid-twentieth century design from the US West-coast.

4 CONCLUSIONS AND PERSPECTIVES

In this deliverable, we have reported upon key design investigations leading to the definition of design rules as manifested in a digital workflow, to enable the proposition of Fungal Architectures. We have developed a novel construction concept based on the combination of two of humanities oldest, and most enduring, technologies - fermentation and weaving. The construction concept has been nuanced and revised in light of various experimental results and findings conducted throughout the project. This has led to a strong association between the possibilities of Kagome weaving to approximate arbitrary design geometries and the sensing capabilities of living mycelium. We have hypothesised that macro morphology of a Fungal Architecture could be considered as the basis for a computationally performing architecture, specific to location. We identify this as a promising area for future research.

We have also shown that the construction concept can be used to achieve such computational goals, mediated through shape and in dialogue with many other architectural design factors, including spatial organisation, structural performance and material specification. The design rules developed for Fungal Architectures extend beyond a material focus and establish a framework for generating novel architectural designs that embody unique spatial and tectonic qualities. Furthermore, the design rules embody a construction approach that is grounded in fundamental and accessible technologies with low barriers to entry. This not only extends these technologies into novel domains of application, but also underscores an effort to make an accessible and inclusive architecture.

We have also reported upon investigations for interfacing living mycelium in ways to have direct and real-time architectural consequence, through a lighting system directly linked to living mycelium. This is the first time, as far as we are aware, that a living mycelium interface has been linked to provide direct internal modification of ambient lighting within a use-inspired architectural context.

In conclusion, we see a promising future architectural (but necessarily and essentially cross-disciplinary) research area emerging in which the deeper integration of living mycelium - or other living materials - act as fully integrated building management systems. A key opportunity here would be consider biological linkages growing to expand beyond the boundary of individual buildings and spaces, to create an inherently linked network at ecological scale and with ecological imperatives.

Further research should also focus on engaging with communities to test transfers of knowledge in the making of weaves and MBC materials and the know-how for the exploitation of local resources to create vernacular architectural expressions that are ecologically localised.

REFERENCES

- [1] Phil Ayres et al. “Meshing with Kagome Singularities: Topology adjustment for representing weaves with double curvature”. In: *Advances in Architectural Geometry*. Ponts Chaussées. 2021, pp. 188–207.
- [2] Adrien Rigobello and Phil Ayres. “Design strategies for mycelium-based composites”. In: *Fungi and Fungal Products in Human welfare and Biotechnology*. Springer, 2023, pp. 605–635.
- [3] Adrien Rigobello and Phil Ayres. “Compressive behaviour of anisotropic mycelium-based composites”. In: *Scientific Reports* 12.1 (2022), p. 6846.
- [4] Adrien Rigobello, Claudia Colmo and Phil Ayres. “Effect of composition strategies on mycelium-based composites flexural behaviour”. In: *Biomimetics* 7.2 (2022), p. 53.
- [5] Felix Heisel et al. “Design, Cultivation and application of load-bearing mycelium components: The MycoTree at the 2017 seoul biennale of architecture and urbanism”. In: *Int. J. Sustain. Energy* 6.1 (2017), p. 18.
- [6] Felix Heisel et al. “Design of a load-bearing mycelium structure through informed structural engineering”. In: *Proceedings of the World Congress on Sustainable Technologies (WCST)*. 2017, pp. 1–5.
- [7] Felix Heisel and Dirk E Hebel. “Pioneering construction materials through prototypological research”. In: *Biomimetics* 4.3 (2019), p. 56.
- [8] Andrew Adamatzky, Antoni Gandia and Alessandro Chiolerio. “Towards fungal sensing skin”. In: *Fungal biology and biotechnology* 8.1 (2021), pp. 1–7.
- [9] Rolf Pfeifer and Gabriel Gómez. “Morphological Computation-Connecting Brain, Body, and Environment.” In: *Creating brain-like intelligence* 5436 (2009), pp. 66–83.
- [10] Helmut Hauser et al. “Towards a theoretical foundation for morphological computation with compliant bodies”. In: *Biological cybernetics* 105 (2011), pp. 355–370.
- [11] Rudolf M Fuchslin et al. “Morphological computation and morphological control: steps toward a formal theory and applications”. In: *Artificial life* 19.1 (2013), pp. 9–34.
- [12] Keyan Zahedi and Nihat Ay. “Quantifying morphological computation”. In: *Entropy* 15.5 (2013), pp. 1887–1915.